

IOWA STATE UNIVERSITY · GO-ECPE · MAY 19, 2026

What Makes a Powerful Engineering Resume?

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YOUR HOST

Who Am I — and Why Does This Matter?

BACKGROUND

 **M.Eng. + B.S. Mechanical Engineering**
Iowa State University — *your school*

 **10 years engineering career · 8 at John Deere**
4 promotions · 4 discipline pivots

 **Mech → Embedded Software → Systems → Eng. Leadership**
Navigated ~~every~~ transition from inside the same company

 **Both sides of the hiring table**
Applied to jobs · Hired engineers as manager · interviewed candidates

 **Systems Engineering Certificate**
California Institute of Technology (Caltech)

"I am not an HR person. I am not a recruiter. I am an engineer who figured out the game — and now I teach it from that side of the table."

CLIENT RESULTS

 **Karen B. — Fortune 100 → SpaceX, Google, GE & Bridgestone offers**
Long-tenured engineer. Rebuilt positioning, resume & interviews. Multiple top-tier offers.

 **Gavin P. — New grad → Fortune 500 Software Engineer**
No connections at target company. Built interview skills from scratch. Landed the role.

 **Jonah B. — Entry-level MechE → Software Engineer**
Internal pivot at Fortune 100. "0% chance" — became Eng II in 18 months.

 **Mukul K. — Career fair rejection → top-tier engineering role**
Rebuilt resume & interview stories after not getting the internship. Landed first great role in Texas.

 **Mohammed & Osaid — 5/5 experience · 5/5 value**
Completed the Engineering Career Accelerator. Both gave top ratings on every dimension.

Disclaimers: 1) What has worked for me.
2) Science is art.

TODAY'S SESSION

What We're Covering Today

- 1 Why great engineers don't hear back — the actual root cause 5 min
- 2 The 4-Part Impact Bullet Builder™ Framework — A · M · H · W 10 min
- 3 Your Group's Report Card — Strengths & Gaps from all 8 audits 8 min
- 4 Your Survey Responses — Pain points you submitted 2 min
- 5 ● Live Demo — 3 Bullet Transformations (interactive) 10 min
- 6 The 6 Highest-Impact Changes for This Cohort 5 min
- 7 Open Q&A 10 min

THE CORE ISSUE

The experience is real.
How it's packaged and presented is a problem.

"The gap between ★★ and ★★★ is not a gap in experience or ability.
It's a gap in translation — and translation is a learnable skill."

50/100

GROUP AVERAGE

39

LOWEST SCORE

66

HIGHEST SCORE

7/8

LANDED IN ★★

Appetizer < Impact
Value > Future ↓

THE ROOT CAUSE

Why Great Engineers Get Ignored

Description
Responsibility



specific, acute

1) X

Bullets read like ~~job duties~~, not accomplishments

"Responsible for managing the test pipeline" — a duty. Hiring managers see hundreds. They all sound the same.

2) X

No numbers backing up any claims

"Improved efficiency" means nothing. By how much? Over how many nodes? Across what time period?

3) X

No detail on HOW you actually did it

Your tools and methods should appear inside the bullet — not just in a skills section at the bottom of the page.

4) X

The engineering impact is nowhere to be found

Why did the work matter? What did it enable? If a hiring engineer can't tell — the resume goes in the pile.

WHAT YOU'RE EXPERIENCING

The Symptom You All Described

5 of 8 of you wrote the same thing in your survey:

"Not hearing back at all."



This isn't a competition problem.

It's a **communication problem**.

The work is real. The results just aren't making it onto the page.

THE FRAMEWORK

The Impact Bullet Builder™

A 4-part formula for turning engineering and research work into recruiter-readable results

A

Accomplishment

What did you DO — with a result?

M

Metric

A specific number with a unit

H

How / Method

Tool, algorithm, or framework
— named in the bullet

W

Why / Impact

What does it enable, replace,
or make possible?

Formula used by engineers who landed roles at Google · SpaceX · John Deere · Ford Motor Company · GE

A
Accomplishment

specific

Something you **DID with a result** — not a duty, not a responsibility, not an ongoing standing task.

THE TEST

"Could someone else with the same job title have written this exact bullet?"

If yes → it's describing the role, not you. Rewrite it.

★ DUTY

"Responsible for maintaining integration test infrastructure."

★★★ ACCOMPLISHMENT

"Redesigned the integration test pipeline — reducing suite runtime by 62% and eliminating flaky failures across 14 CI environments."

M

Metric

A specific number that gives the claim a floor to stand on.

WHAT COUNTS FOR EE/CE/CS RESEARCH

% · dB · ms · accuracy · F1 score · BLEU score

dataset size · node count · throughput (Mbps)

PCE (%) · XRD crystallinity index · entanglement fidelity

kW · publication venue · benchmark rank

NOT METRICS

"several" · "multiple" · "significant" · "improved" · "various" ·
"many"

💡 The number is already in your lab notebook, code output, or poster file.
It just hasn't made it to your resume yet.

1) proof $\Rightarrow$ you did it

2) BS meter $\downarrow$

did the thing $\rightarrow$ knows the thing

processes, people,
methods, major hurdles

✓ ✓ ✓
Name the specific **tool, algorithm, or framework** inside the
bullet text — not in the role header, not in the skills section.

H

How / Method

EXAMPLES

- ✓ "using GAMS and CPLEX"
- ✓ "via a PyTorch-PennyLane hybrid workflow"
- ✓ "with TensorRT FP16 quantization"
- ✓ "using the MUSIC direction-finding algorithm"
- ✓ "via SEM and XRD characterization"

Your group scored **6.75/10** here — above average. The instinct
to name tools in context is right. Keep doing it.

- 1) proof
- 2) BS meter.

3) helps w/ skills $\Rightarrow$ python $\Rightarrow$ context

\$, delivery, ---

★ **W**
Why / Impact

- 1) proof
- 2) BS meter
- 3) who you are

"so what?"

Connect the result to an **engineering purpose**. What does it enable? What does it replace? What becomes possible because of it?

X NOT WHY

"Improving performance" / "Enhancing system efficiency"

✓ WHY — WITH ENGINEERING PURPOSE

- ✓ "Enabling real-time spectrum allocation without centralized coordination"
- ✓ "Enabling planning studies at a fraction of prior computational cost"
- ✓ "Enabling deployment in clinical settings with limited labeled data"

PUT IT TOGETHER

The Full Formula

[Strong Verb] + [What You Built] using [Tool / Method] –
achieving [Metric], enabling [Why / Impact]

★ BEFORE — 1 COMPONENT (WEAK A, NO M, NO H, NO W)

"Implemented a graph-based algorithm for interference detection in dynamic spectrum sharing environments."

★★★ AFTER — ALL 4 COMPONENTS

"Developed a GNN-based spectrum sharing algorithm in PyTorch — reducing co-channel interference by 23 dB across a 64-node simulation, enabling real-time dynamic allocation without centralized coordination."

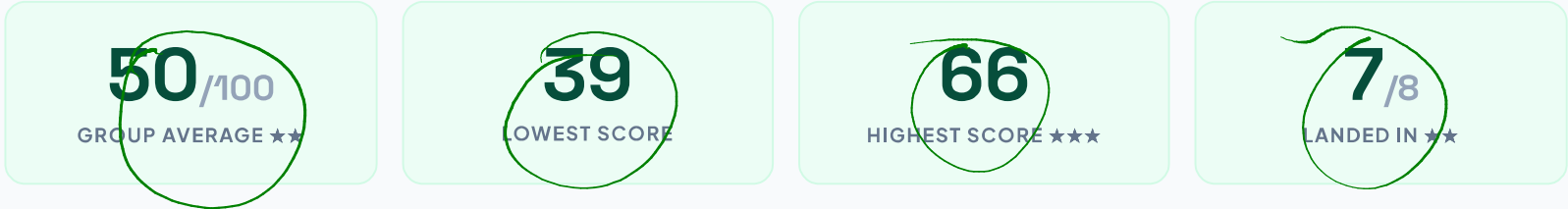
★ = 1 component

★★ = 2–3 components

★★★ = all four

YOUR GROUP'S REPORT

Your Group's Report Card

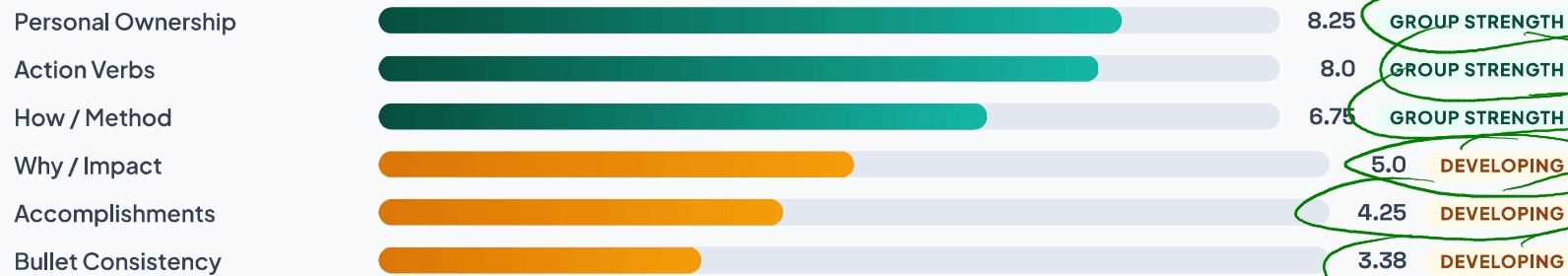


CANDIDATE (ANONYMIZED)	SCORE	TIER	RESEARCH AREA
Candidate A — Ph.D., Computer Engineering	39	★★	LLM/VLM Evaluation · Medical AI
Candidate B — Ph.D., Computer Engineering	41	★★	Wireless Comms · Spectrum Sharing · GNN
Candidate C — Ph.D., Electrical Engineering	41	★★	Perovskite Solar Cell · Microelectronics
Candidate D — Ph.D., Electrical Engineering	45	★★	Power Systems Planning · Renewable Energy
Candidate E — Ph.D., Computer Science	51	★★	AI/Robotics · Computer Vision
Candidate F — M.S., Computer Engineering	56	★★	Quantum Security · Intrusion Detection
Candidate G — Ph.D., Electrical Engineering	59	★★	RF/MW · IC Design · PLL Design
Candidate H — Ph.D., Computer Science	66	★★★	AI Engineering · LLM Systems · RAG

CRITERION ANALYSIS

What Your Group Is Doing Well

Group averages per criterion — scored across all 8 resumes (max 10.0 per criterion)



✓ You own your work with first-person active verbs. Tools appear inside bullets, not just in skills lists. The foundation is strong — these gaps are specific and fixable.

CRITERION ANALYSIS

What's Holding Scores Back

Two criteria account for most of the distance between ★★ and ★★★

Metrics / Quantification



2.88 PRIMARY GAP

Problem Context



2.38 PRIMARY GAP

Metrics — 2.88/10 · 6 of 8 scored 1–2

Every domain here produces measurable outputs: PCE%, dB, F1, ms, XRD index, entanglement fidelity, Mbps. These numbers exist — in lab notebooks, code output, poster files. Not one is required to be invented. They just aren't written down.

Problem Context — 2.38/10 · Lowest criterion

Almost no bullets establish what existed before, what gap motivated the work, or what the before-state was. Results appear without a baseline. The best bullet in all 8 resumes: *"Reduced ~90,000-bus system to ~900 buses..."* — it works because it shows before and after.

THE DEFINING ISSUE

The PhD Research Problem

8 of 8 candidates: PhD research is underdeveloped or invisible on the resume.

COMMON PATTERNS FOUND

- Research described in one line under "Education → Research Focus"
- Present-participle verbs: "Exploring..." / "Working on..." / "Investigating..."
- Publications listed separately, never tied to the bullets that produced them
- Research written for other grad students, not hiring engineers

WHAT TO DO INSTEAD

"Your RA role is your most important credential.
Write it like your most important job."

Every research bullet needs 3 things:

- ① Named approach — what you're building or using
- ② Scope indicator — nodes, scenarios, models, measurements
- ③ Direction or preliminary result — what it targets or what you've found

Bullets

WHAT YOU TOLD ME

Your Survey Responses

"I am not hearing back at all."

— Hadia · Wireless Eng. Co-op applicant

"I struggle to translate PhD projects into clear, concise resume bullets that match what recruiters are looking for."

— Himani · AI/ML, PhD

"PhD research-level projects are often irrelevant for industry — how do you cater to this problem?"

— Zaid · Robotics, PhD

"Not hearing back, or getting rejected due to high competition."

— Zohreh · Power Systems internship applicant

"Not hearing back from employers after applying — not even getting a reason."

— Sibtain · AI Engineer (LLM), PhD

All of these trace to the same root cause: **the results aren't stated on the page.**
Not a competition problem. Not an experience problem. A translation problem.

LIVE DEMO

10–Minute Transformation Exercise

3 bullets. Patterns I found across your group's resumes. Let's fix them live.

How This Works

- Step 1: I show the BEFORE bullet
- Step 2: We diagnose — what's missing?
- Step 3: Press → to reveal the AFTER, tagged by A · M · H · W

The 3 Domains

- Demo 1 — Wireless / RF Research
- Demo 2 — AI / ML Pipeline
- Demo 3 — Power Systems / Circuit Design

Press → on each demo slide to reveal the After panel

Spectrum Sharing & Interference Detection

BEFORE — ★ (1 OF 4 COMPONENTS)

"Implemented a graph-based algorithm for interference detection in dynamic spectrum sharing environments."

DIAGNOSE — WHAT'S MISSING?

Algorithm named (graph-based) ✓. But: **no metric** (how much interference reduced?), **no scale** (how many nodes?), **no Why** (what does this enable for the system?)

AFTER — ★★★ (ALL 4 COMPONENTS)

*"Developed a **GNN-based spectrum sharing algorithm in PyTorch** — reducing co-channel interference by **23 dB** across a **64-node simulation**, enabling **real-time dynamic allocation without centralized coordination**."*

A — Developed · reduced interference

M — 23 dB · 64-node simulation

H — GNN-based · PyTorch

W — Real-time allocation · no central coord

DEMO #2 — AI / ML PIPELINE

Medical Imaging & Deep Learning Ensemble

BEFORE — ★ (PARTIAL H, NO M, NO A RESULT, NO W)

"Built an ML-DL hybrid pipeline using multiple deep learning architectures and ensemble learning for medical image classification."

DIAGNOSE — WHAT'S MISSING?

Says "multiple architectures" but doesn't name them. **No accuracy figure**, no dataset size, no baseline comparison, no condition named, no publication link. The IEEE poster for this exact work listed the accuracy figures. None of it made it to the resume.

AFTER — ★★★ (ALL 4 COMPONENTS)

*"Implemented a **VGG19/ResNet50/EfficientNetB0 ensemble** for diabetic retinopathy detection — achieving **94.2% accuracy on 3,500 test images**, outperforming the single-model baseline by **8.4%**, with findings **submitted to IEEE BHI 2025**."*

A — Implemented · outperformed baseline

M — 94.2% accuracy · 3,500 images · 8.4% gain

H — VGG19/ResNet50/EfficientNetB0 named

W — IEEE BHI submission · clinical detection

DEMO #3 — POWER SYSTEMS / CIRCUIT DESIGN

Power Converter & Grid Integration

BEFORE — ★ (VAGUE A · NO M · NO H · WEAK W)

"Designed and simulated a power converter circuit to improve system efficiency for grid applications."

DIAGNOSE — WHAT'S MISSING?

No tool named (MATLAB/Simulink? LTspice?), **no efficiency figure**, **no power rating**, **no baseline** ("improve" from what?), **no application specificity** ("grid" tells nothing about scale or use case).

AFTER — ★★★ (ALL 4 COMPONENTS)

*"Designed a bidirectional DC-DC converter in **MATLAB/Simulink** — achieving **95.3% efficiency at 500 kW peak load** and reducing switching losses by **18%** vs. the prior design, enabling **integration with a 100-node IEEE test bus for planning-study validation**."*

A — Designed · achieving, reducing

M — 95.3% · 500 kW · 18% reduction · 100-node

H — MATLAB/Simulink · bidirectional DC-DC

W — IEEE test bus · planning-study validation

ACTION PLAN

The 6 Highest-Impact Changes



1

Add 1 metric to every research bullet

Numbers are in your lab notebooks, code output, and poster files. They don't need to be invented — they've already been measured.



2

Write your PhD research section like your most important job

Named approach + scope indicator + direction or preliminary result. That's all it takes.



3

Link your publications to the bullets that produced them

Four words: *"with findings published at [venue]."* Do this for every paper you have.



4

Add a "before" state to each major contribution

Give the baseline. "Improved X" means nothing without knowing what X was beforehand.



5

Replace "Exploring / Working on / Investigating"

Convert to past tense with a stated output — or add "with initial results showing [X]" to any ongoing work.



6

If you have industry experience, write it like it matters — because it does

Dates-only entries are worse than omitting them. Every industry role is a differentiator for PhD applicants.

LIVE Q&A

Questions You Submitted

- Q1** How do I turn ongoing PhD research into strong, industry-style resume bullets — even before it's published?
- Q2** Should research papers go above or alongside industry experience? What's the right structure for a PhD resume?
- Q3** How do I tailor my resume for different roles efficiently — without spending 2 hours per application?
- Q4** How do I talk about research experience in interviews vs. traditional engineering behavioral interviews?
- Q5** What do recruiters at hardware, AI, and power systems companies actually look for in graduate engineer resumes?

WHAT COMES NEXT

Take This Further

YOUR INDIVIDUAL AUDIT

Your personal audit report was delivered to the email you submitted. It includes your score, criterion-by-criterion breakdown, and the top-priority bullets to rewrite first.

REFERRAL BONUS

\$200 cash for any club member who refers someone to the Engineering Career Accelerator. Make sure they mention your name at booking.

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THE CORE MESSAGE

**"The research is real.
The publications are real.
The technical depth is real.**

**The gap is the translation —
and that's a learnable skill."**

Mansour Manci
Engineering Career Coach

LaunchToLead.io
Free strategy call available

LinkedIn
[/in/mansourmanci](#)

Thank You — Open Q&A